Int J Adv Manuf Technol (2008) 38:258–267
DOI 10.1007/s00170-007-1106-9

ORIGINAL ARTICLE

Modeling of heat transfer and fluid flow during gas tungsten arc welding of commercial pure aluminum

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Received: 4 March 2007 / Accepted: 21 May 2007 / Published online: 28 July 2007
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Abstract In the present study, the temperature and the velocity fields during gas tungsten arc welding of commercial pure aluminum were simulated using the solution of the equations of conversation of mass, energy and momentum in three dimensions and under steady-state heat transfer and fluid flow conditions. Then, by means of the prediction of temperature and velocity distributions, the weld pool geometry, weld thermal cycles and various solidification parameters were calculated. To verify the modeling results, welding experiments were conducted on two samples with different thicknesses and the geometry of the weld pool was measured. It is found that there is a good agreement between the predicted and the measured results. In addition, dimensional analysis was employed to understand the importance of heat transfer by convection and the roles of various driving forces in the weld pool. It is observed that the molten metal convection strongly affects on the weld pool geometry. Also the predictions make it possible to estimate the morphology and the scale of the solidified structure through solidification parameter (G/R). The result show that as the net heat input increases, the importance of convection becomes higher and the value of G/R at the weld pool centerline increases.

Keywords Modeling · Heat transfer · Fluid flow · Gas tungsten arc (GTA) welding · Commercial pure aluminum · Solidification parameter

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1 Introduction

Since the temperature and velocity distributions of the molten metal as wells the cooling rate after welding operation affect the weld geometry, the microstructure, and the mechanical properties of weld zone, there has been a significant interest in the quantitative representation of heat transfer and fluid flow phenomena in the weld pools. On the other hand, because of the small size of the weld pool, the presence of plasma in the vicinity of the weld pool and rapid changes of temperature in arc spot welds, physical measurements of important parameters, such as temperature and velocity fields, solidification rate, and thermal gradient in the weld pool are extremely difficult and experimental study of the flow conditions in the weld pools has been limited to the measurement of the surface velocities. Therefore, in recent decades, numerical calculations of heat transfer and fluid flow have been utilized to understand various weld characteristics that cannot be obtained otherwise.

In the first attempts, significant progress has been made in the solution of the equations of conservation of mass, momentum and energy in fusion welding with a stationary heat source [1–3]. However, in practice, the heat source moves with a certain velocity. Thus, several investigations

of fluid flow and heat transfer in welding with moving heat source have been reported [4–7].

The three-dimensional (3D) comprehensive numerical heat transfer and fluid flow model used in the calculations has been extensively validated for the welding of pure iron [3, 8], steels [9–20], aluminum alloys (5182 and Al-Cu) [21–23], and titanium alloys [24, 25]. However, most of these studies were concerned with iron and its alloys while the heat transfer and fluid flow in the commercial pure aluminum have not received much attention in the literature.

Numerical calculations of heat transfer and fluid flow in welding have enabled accurate quantitative calculations of thermal cycles and fusion zone geometry [6, 26, 27]. In many simple systems, the computed thermal cycles have been used to quantitatively understand weld metal phase composition [28], grain structure [24], inclusion structure [9, 12] and weld metal composition changes, owing to both vaporization of alloying elements [21] and dissolution of gases [29, 30]. Capabilities to quantitatively understand geometry, composition and structure of welds in simple systems have provided hope that one day welding engineers may be able to use numerical models to tailor weldment characteristics according to specifications. In reality, the numerical heat transfer and fluid flow codes for fusion welding have so far been used mostly as a research tool [10, 11, 22] rather than as a tool for design and manufacturing in the industry, but application of commercial code such as FLUENT can applicably aid to reach this goal.

In this study, the heat transfer and fluid flow gas tungsten arc (GTA) welds of commercial pure aluminum were simulated using a well-tested, three-dimensional, steady state, numerical heat transfer and fluid flow model. The model was applied to understand the temperature field and the weld dimensions for various GTA welding conditions. For the accuracy of the calculation, a very fine grid system was used. Verification of the model was performed through comparing the calculated results with metallographic weld cross-sections. Dimensional analysis was carried out to understand the heat transfer mechanism and the importance of the various driving forces for the liquid pool convection. Also, several important solidification parameters, including solidification rate (R) and temperature gradient (G) at the weld centerline, were calculated. These parameters may be used for determination of the solidification morphology and the scale of the solidification substructure. The effects of the net heat input and welding speed on the heat transfer and fluid flow were analyzed.

2 Mathematical formulation

The following assumptions were made in the present study: 1) The flow is Newtonian and incompressible, in view of the relatively small size of weld pools expected. 2) All of the physical properties of the liquid and solid metals are constant, independent of the temperature, except for surface tension and thermal conductivity. 3) The spatially-distributed heat and current fluxes falling on the free surface have Gaussian characteristic. 4) The density variation in the calculation domain is ignored except for the calculation of the buoyancy force following Bossinesq's approximation. 5) The weld top surface is assumed to be flat.

In practice, the heat source moves with a certain constant velocity. Thus, when viewed in a fixed coordinate system (x', y, z), the welding problem is unsteady. But in the coordinate system that moves with the heat source (x, y, z), the problem becomes steady a short time after the start of welding. Under such conditions, the heat source and molten metal under the heat source are fixed in space, and the material enters and leaves the computational domain at the welding velocity. The material inlet velocity is taken as equal in magnitude to the welding velocity but opposite in direction.

Therefore, the equations of conservation of mass, momentum, and energy are first formulated in the coordinate system (x, y, z) attached to the moving heat source. To treat the convective velocity as the primary unknown in the governing equations, the equations are further modified by subdiving the net velocity into convective and welding velocity components as follows:

$$\boldsymbol{V}' = \boldsymbol{V} + \boldsymbol{U} \tag{1}$$

where $\mathbf{V}'$ is the net velocity at any point with respect to the frame attached to the heat source, $\mathbf{V}$ is the convective component of the velocity, and $\mathbf{U}$ is the welding speed along the positive x direction.

Electromagnetic, surface tension, and buoyancy driving forces were considered for the calculation of weld pool convection. The weld pool boundary was traced using an enthalpy-porosity technique in a fixed Cartesian coordinate system.

Henceforth, the following tensorial notation will be followed for description of the conservation equations:

$$x_1 = x, x_2 = y, x_3 = z.$$

2.1 Governing equations

Using the assumptions stated above, one can represent the circulation of liquid metal in the weld pool by the following linear momentum conservation equation for the jth direction:

$$\rho\frac{\partial(u_i u_j)}{\partial x_i} = -\frac{\partial P}{\partial x_j} + \frac{\partial}{\partial x_i}\left(\mu\frac{\partial u_j}{\partial x_i}\right) + \frac{\partial}{\partial x_i}\left(\mu\frac{\partial u_j}{\partial x_j}\right) \rho U\frac{\partial u_i}{\partial x_i} - C\left(\frac{(1-f_L)^2}{f_L^3 + B}\right)u_j + Sb_j + Se_j, \tag{2}$$

where ρ is density, x_i is the distance along the i=1, 2, and 3 directions, u_j is the velocity component along the j direction, μ is the viscosity, P is the pressure, f_L is the liquid fraction, B is a constant introduced to avoid division by zero, C is a constant that considers mushy zone morphology, U is the welding speed, and Sb_j and Se_j present the electromagnetic and buoyancy source terms, respectively. In Eq. (2), the first term on the right-hand side is the pressure gradient. The second term is the viscosity arising from casting the momentum equation into a general form. The fifth term describes the frictional dissipation in the mushy zone according to the Carman-Kozeny equation for flow through a porous media. The calculation of the electromagnetic and buoyancy source terms is expressed in the literature [1, 6, 31].

The pressure field was obtained by solving following continuity equation simultaneously with the momentum equation:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0. \tag{3}$$

The total enthalpy H is represented by a sum of sensible heat h and latent heat content ΔH, i.e., $H = h + \Delta H$, where $h = \int C_p dT$, C_p is the specific heat, T is the temperature, $\Delta H = f_L L$, L is the latent heat of fusion, and the liquid fraction f_L is assumed to vary linearly with temperature in the mushy zone:

$$f_L = \begin{cases} 1, T > T_L \\ \dfrac{T - T_s}{T_L - T_s}, T_s \le T \le T_L \\ 0, T < T_s \end{cases} \tag{4}$$

where T_L and T_S are the liquidus and the solidus temperatures, respectively. The thermal energy transport in the weld workpiece can be expressed by the following modified energy equation:

$$\rho \frac{\partial(u_i h)}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\frac{k}{C_p}\frac{\partial h}{\partial x_i}\right) + \rho \frac{\partial(u_i \Delta H)}{\partial x_i} + \rho U \frac{\partial \Delta H}{\partial x_i} + \rho U \frac{\partial h}{\partial x_i} \tag{5}$$

where k is the thermal conductivity.

2.2 Boundary conditions

A 3D Cartesian coordinate system is used in the calculation, while only half of the workpiece is considered since the weld is symmetrical about the weld center line. Figure 1 is a schematic plot showing the boundary conditions. These boundary conditions are further discussed as follows.

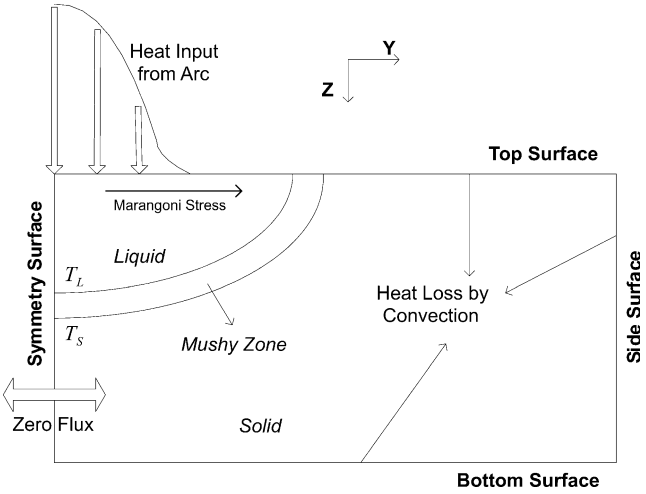


Fig. 1 Boundary conditions used in calculation of temperature and velocity fields

2.2.1 Top surface

The velocity boundary conditions is given as

$$\begin{aligned} \mu \frac{\partial u_1}{\partial z} &= f_L \frac{d\gamma}{dT}\frac{\partial T}{\partial x_1} \\ \mu \frac{\partial u_2}{\partial z} &= f_L \frac{d\gamma}{dT}\frac{\partial T}{\partial x_2} \\ u_3 &= 0 \end{aligned} \tag{6}$$

where u_1, u_2, and u_3 are the velocity components along the x, y, and z directions, respectively, and $d\gamma/dT$ is the temperature coefficient of surface tension. As shown in this equation, the u_1 and u_2 velocities are determined from the Marangoni effect. The u_3 velocity is equal to zero since there is no flow of liquid metal perpendicular to the pool top surface.

The heat flux at the top surface is given as

$$k \frac{\partial T}{\partial x_3}\Big|_{top} = \frac{fQ\eta}{r_b^2} \exp\left(-\frac{f(x^2 + y^2)}{r_b^2}\right) - h(T - T_a) \tag{7}$$

where r_b is the heat distribution parameter, f is the power distribution factor, Q is the total arc power, η is the power efficiency, h is the heat transfer coefficient, and T_a is the ambient temperature. The first term on the right-hand side of Eq. (7) is the heat input from the heat source, defined by a Gaussian heat distribution. The arc power distribution factor "f" determines the nature of the Gaussian heat distribution pattern. The distribution is rather diffused for

low values of "f" such as 0.5 and more focused for higher values of d such as 3.0. The second term represents the heat loss by convection.

2.2.2 Symmetric surface

The boundary condition are defined as zero across the symmetric surface as

$$\frac{\partial u_1}{\partial x_2} = 0, u_2 = 0, \frac{\partial u_3}{\partial x_2} = 0, \quad \frac{\partial h}{\partial x_2} = 0. \tag{8}$$

2.2.3 Other surface

At all other surfaces, the heat losses by convection and the velocities are set to be zero.

2.3 Discretization of governing equations and grid spacing

The detailed method of discretizing the governing equations is available in the literature [6]. A brief description is presented here to highlight the salient features. The governing equations are discretized using the control volume method [32], where the computational domain is divided into small rectangular control volumes. A scalar grid point is located at the center of each control volume, storing the values of scalar variable such as pressure and enthalpy. Velocity components are stored at separate locations, staggered with respect to scalar locations, to ensure the stability of numerical calculation. Thus the control volumes for scalars are different from those for the vectors. Discretized equations for a variable are formulated by integrating the corresponding governing equation over the control volumes. After Discretization, the conservation equation for a general variable ϕ such as velocity or enthalpy at a cell P can be written as [32–34]:

$$a_P \phi_P = \sum_{nb} (a_{nb} \phi_{nb}) + b \tag{9}$$

here a_P is the center coefficient, a_{nb} are the influence coefficients for the neighboring cells, and b is the contribution of the constant part of source term S_c in $S = S_c + S_P \phi_P$ and of the boundary conditions. Power-law scheme is employed to evaluate the combined convection/diffusion coefficients a_{nb}[32–34]. In Eq. (9),

$$a_P = \sum_{nb} a_{nb} - S_P \tag{10}$$

To numerically solve the governing equations with the associated source terms, a general computational fluid dynamics (CFD) computer program, FLUENT code was used. The material properties and workpiece information are input data, employed in the calculation of fluid flow and heat transfer are summarized in Table 1 [35].

The velocity-pressure coupling in the discretized momentum equations is handled using the semi-implicit method for pressure-linked equations (SIMPLE) algorithm at the each iteration[32–34].

Accurate calculation of weld temperature and velocity fields, and the temperature gradients requires the use of a very fine grid system. A typical grid system used in the present study contained 143 × 73 × 50 grid points, and the corresponding computational domain had dimensions of 100 mm in length, 50 mm in width, and 5 mm in depth. Spatially non-uniform grids were used for maximum resolution of variables. A finer grid spacing was used near the heat source. The minimum grid spacings along the x and z directions were about 50 and 20 μm, respectively.

2.4 Convergence criteria

The residual R^{ϕ} computed by FLUENT's segregated solver is the imbalance in Eq. (9) summed over all the computational cells P. It may be written as

$$R^{\phi} = \frac{\sum_{cellsP} \left| \sum_{nb} (a_{nb} \phi_{nb}) + b - a_P \phi_P \right|}{\sum_{cellsP} |a_P \phi_P|} \tag{11}$$

The residual values should be usually very small when a converged solution is obtained. In the present study, the convergence was reached when $R^{\phi} \leq 10^{-4}$ for the energy equation and $R^{\phi} \leq 10^{-3}$ for the momentum equations.

Table 1 Physical properties of commercial pure aluminum used in the calculations [35]

Physical property	Value
Liquid temperature, T_L (K)	930
Solid temperature, T_S (K)	916
Density of liquid metal, ρ (kg/m^3)	2700
Viscosity of liquid, μ (kg/m s)	0.0014
Thermal conductivity of solid, k_S (J/m s K)	222
Thermal conductivity of liquid, k_L (J/m s K)	108
Specific heat of solid, C_{PS} (J/kg K)	904
Specific heat of liquid, C_{PL} (J/kg K)	904
Latent heat of melting, ΔH (J/kg)	3.95×10^5
Temperature coefficient of surface tension, $d\gamma/dT$ (N/m K)	-0.152×10^{-3}
Coefficient of thermal expansion, β (K^{-1})	2.55×10^{-5}
Magnetic permeability, μ_m (N/A^2)	1.26×10^{-6}

3 Experimental procedure

Commercial pure aluminum plates with the thicknesses of 5 mm and 10 mm were used in this investigation. The composition of the as-received plate was (by wt%): 0.08 Si, 0.29 Fe, 0.005 Cu, 0.002 Mn, 0.002 Mg, 0.06 Zn, 0.003 Ti, 0.001 Cr, and 0.002 Ni. GTA welds were performed on the aluminum plates using a 300 A direct constant current welding power supply with electrode negative polarity. The welding electrode was made of W-2%Th measuring 2.4 and 3.2 mm in diameter. High Purity (99.99%) argon was used as welding gas. The samples were machined out of the plates and welding experiments were conducted under different conditions as listed in Table 2.

4 Results and discussion

4.1 Temperature and velocity fields

The experimentally determined weld pool cross-sections are compared with the corresponding calculated geometries in Fig. 2 (a) and (b). It is observed that the computed weld pool geometry and dimensions agree well with the experimental results. In these figures, the mushy zone, i.e., the solid and liquid two phase region is presented with the distinct color. Note that the mushy zone is determined as the region having temperature between the solidus (916 K) and liquidus (930 K) isotherms.

The net heat input, H_n, is defined as the amount of heat absorbed per unit length of weld from a traveling heat source, i.e., the ratio of absorbed power to the welding speed

$$H_n = \frac{\eta Q}{V} \tag{12}$$

where Q is the power and equal to multiplication of arc current and voltage, V is the welding speed and η has been defined earlier. The calculated values of weld pool depth and half width pool for various heat inputs are shown in Fig. 3. As it is observed, different values of H_n were obtained by changing the welding speed. In addition, the weld pool depth of the sample 1 is higher and the weld pool width is less than the sample 2. Thus, a wide and shallow weld pool is produced due to the higher thickness of the sample 2. Figure 4 indicates the variations in calculated peak temperature and maximum velocity as the net heat input increases by decreasing the welding speed. The weld pools of the sample 1 have higher peak temperatures and liquid metal velocities. The peak temperature varies from 1495 K to 1746 K for the sample 1 and 1556 K to 1802 K for the sample 2 with the net heat input. Also the maximum velocity alters from 1.253 m/s to 1.445 m/s for the sample 1 and 1.225 m/s to 1.420 m/s for the sample 2.

Table 2 Conditions of GTA welded samples

Variables	Sample 1	Sample 2
Thickness (mm)	5	10
Current (A)	110	150
Voltage (V)	15	17
Welding speed (mm/s)	2.7	7.1

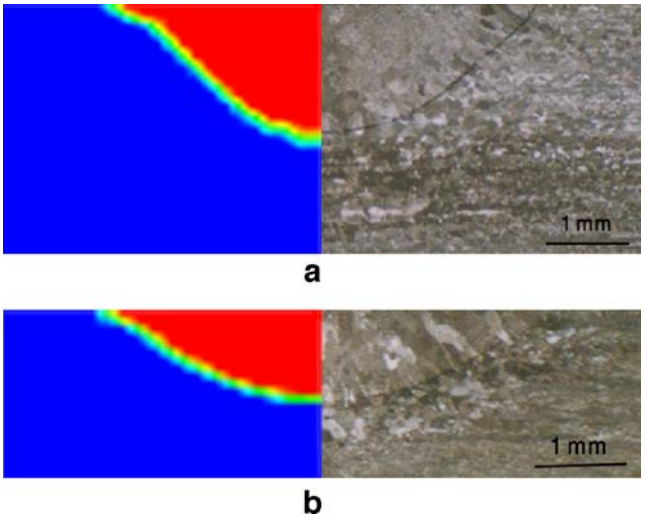


Fig. 2 Experimental and calculated weld pool cross-sections for GTA welds obtained using the conditions of the (**a**) Sample 1; (**b**) Sample 2

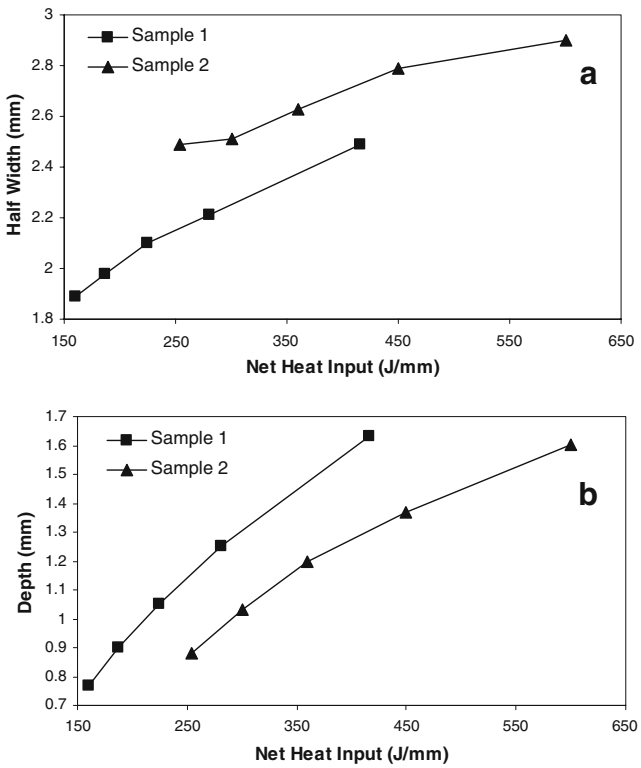


Fig. 3 Effects of heat input on (**a**) the half pool width and (**b**) the weld pool depth for the samples 1 and 2 by changing in the net heat input

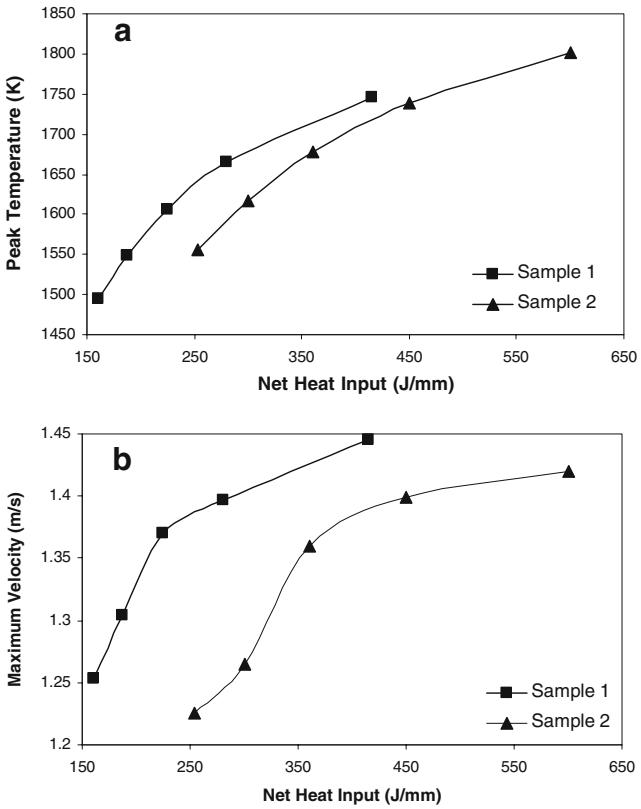


Fig. 4 The variations in (**a**) the peak temperature and (**b**) the maximum velocity with the changes of the net heat input of the samples 1 and 2

Figure 5 shows the computed 3D temperature and velocity fields for a weld pool obtain using the conditions of the sample 1. For clarification; only a half of weld is shown. The liquid metal motion in the weld pool is driven mainly by the surface tension and electromagnetic forces, and, to a much lesser extent, by the buoyancy force. The relative importance of these driving forces is quantitatively examined in a subsequent section. Because the temperature coefficient of surface tension $d\gamma/dT$ has a negative value, the surface tension force drives the liquid metal from the center to the periphery at the top surface of the weld pool.

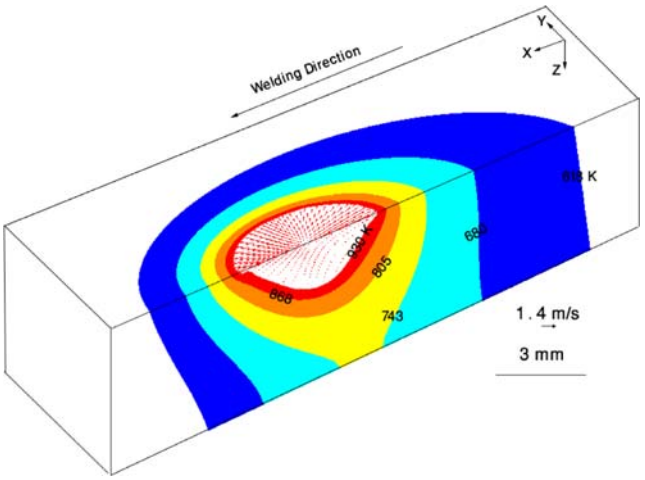


Fig. 5 Calculated temperature and velocity fields in three dimensions made by the conditions of the sample 1

4.2 Weld thermal cycles

The temperature-time data at different locations in the weld metal plays an important role in determining weld metal microstructure and properties. For the steady state temperature field, obtained from the solution of the transformed momentum and enthalpy equation, temperature as function of time at different locations (x, y, z) can be calculated by the following relation [6, 20, 28]:

$$T(x,y,z,t_2) = \frac{T(\xi_2,y,z) - T(\xi_1,y,z)}{\xi_2 - \xi_1} U(t_2 - t_1) + T(x,y,z,t_1) \quad (13)$$

where $T(\xi_2, y, z)$ and $T(\xi_1, y, z)$ are the steady state temperatures at coordinates (ξ_2, y, z) and (ξ_1, y, z), respectively; $(\xi_2\text{-}\xi_1)$ is the distance traveled by the beam in time $(t_2\text{-}t_1)$; and $T(x, y, z, t_1)$ and $T(x, y, z, t_2)$ are the temperatures at location (x, y ,z) at times t_1 and t_2, respectively.

Figure 6 shows the computed thermal cycles at the weld centerline for the sample 2 with different welding speed. It is seen that the peak temperature increases but heating and cooling rates decreases with decreasing the welding speed. Also heating rates is higher than cooling rates.

In Fig. 7, the calculated thermal cycles at different monitoring locations are represented. At locations equidistant from the heat source, the temperatures are higher along y direction (perpendicular to the welding direction) than those along the vertical z direction. This behavior originates

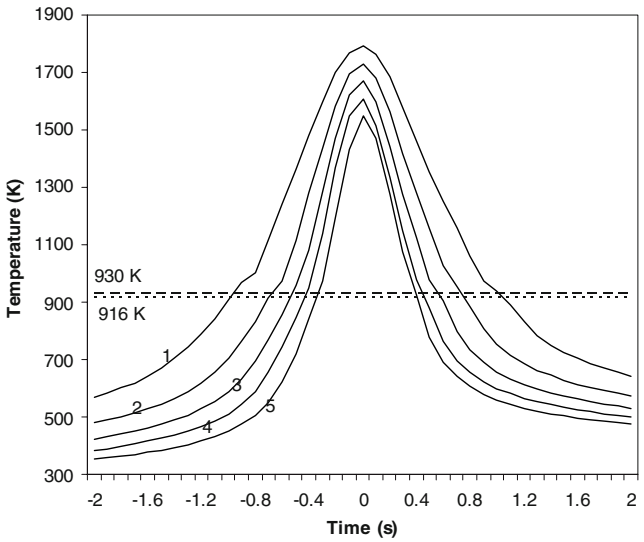


Fig. 6 Calculated thermal cycles for GTA welds obtained using condition of the sample 2 at the weld centerline for different welding speeds: (1) 3 mm/s; (2) 4 mm/s; (3) 5 mm/s; (4) 6 mm/s; (5) 7.1 mm/s

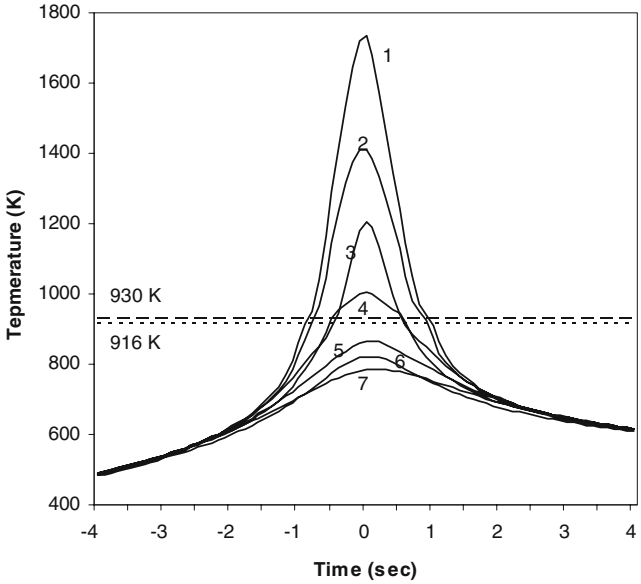


Fig. 7 Calculated thermal cycles at different locations in the sample 1. Distance from weld centerline: (1) y=0 mm, z=0 mm; (2) y=1 mm, z=0 mm; (3) y=0 mm, z=1 mm; (4) y=2 mm, z=0 mm; (5) y=0 mm, z=2 mm; (6) y=3 mm, z=0 mm; (7) y=0 mm, z=3 mm

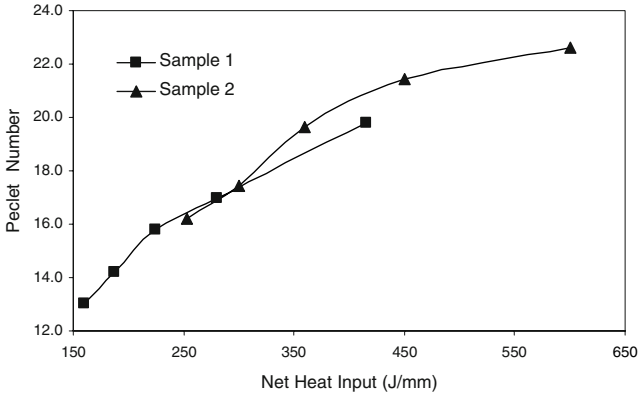


Fig. 8 Calculated Peclet number as a function of the net heat input for the samples 1 and 2

from the specific weld pool geometry that results from the surface-tension driven flow. In other words, the temperature gradient along z direction is higher than that along y direction because of the wide and shallow weld pool. From Fig. 7, it can also be seen that during both the heating and cooling periods, there are inflection points at liquidus temperature in the enthalpies of the thermal cycle curve. This feature results from the differences in the enthalpies of the solid and the liquid metals. The same behavior has been observed in steel elsewhere [10].

4.3 Estimation of scaling using dimensionless numbers

The Peclet number, Pe, is defined by Eq. (14). This number is used to evaluate the relative importance of heat transfer by conduction and convection in the weld pool.

$$Pe = \frac{u_R \rho C_p L_R}{k} \tag{14}$$

where u_R and L_R are the typical velocity and characteristic length of the weld pool, respectively, and the symbols ρ, C_p, and k have been defined earlier. As shown in Fig. 5, outward flow driven by the surface tension force is dominant in the liquid metal circulation and has a significant effect on modifying the weld pool shape. Hence, u_R is taken as the average velocity and L_R is taken as the pool radius at the top surface of the weld pool. Figure 8 shows computed Peclet number as a function of the heat input for the samples 1 and 2. When Pe is much higher than one, similar the samples 1 and 2, heat is transported mainly by convection and liquid metal circulation in the weld pool markedly affects the weld pool geometry. As shown in the Fig. 8, the Peclet number increases with increasing the net heat input or decreasing the welding speed. It indicates the important role of convection resulting from the increase in both the weld pool size and velocity.

The driving forces for the flow in the weld pool considered in the present study include surface tension, electromagnetic, and buoyancy forces. The relative importance of these driving forces can be estimated using several dimensionless numbers, as indicated in Eqs. (15–17).

The ratio of buoyancy force to viscous force is determined by the Grashof number:

$$Gr = \frac{g\beta L_B^3 \Delta T \rho^2}{\mu^2} \tag{15}$$

where g is the gravitational acceleration, β is the thermal expansion coefficient, ΔT is the temperature difference between the peak pool temperature and solidus temperature, and L_B is a characteristic length for the buoyancy force in the liquid pool, which is approximated by one eighth of the weld width [20].

The magnetic Reynolds number defines the ratio of electromagnetic force to viscous force, and is expressed as

$$Rm = \frac{\rho \mu_m I^2}{4\pi^2 \mu^2} \tag{16}$$

where μ_m is magnetic permeability.

The surface tension Reynolds number is used to describe the ratio of surface tension gradient force to viscous force, and is calculated as

$$Ma = \frac{\rho L_R \Delta T |d\gamma/dT|}{\mu^2} \tag{17}$$

The relative importance of primary driving force can thus be judged by the combination of these dimensionless

Table 3 Dimensionless number computed in the weld pool for the samples 1 and 2

Dimensionless number	Value	
	Sample 1	Sample 2
Grashof. Gr	186.1	130.1
Magnetic Reynolds number, Rm	5.3×10^5	9.9×10^5
Surface tension number, Ma	4.3×10^5	3.2×10^5
R_{SIB}	2.3×10^3	2.9×10^3
R_{MIB}	2.9×10^3	7.6×10^3

numbers. For example, the ratio of surface force to buoyancy force is expressed as

$$R_{S/B} = \frac{Ma}{Gr} \tag{18}$$

while the ratio of electromagnetic force to buoyancy force is given by

$$R_{M/B} = \frac{R_m}{Gr} \tag{19}$$

Using the physical properties listed in Table 1, we summarize the calculated dimensionless numbers in Table 3 for the samples 1 and 2.

Based on the values listed in Table 3, it is expected that the liquid flow is mainly driven by the surface tension and electromagnetic forces and, to a much less extent, by the buoyancy force.

Since the surface tension force is the dominant driving force for convection in the weld pool, the order of magnitude the maximum velocity can be approximated by

$$u_m^{3/2} \approx \frac{d\gamma}{dT}\frac{dT}{dR_W}\frac{R_W^{1/2}}{0.664\rho^{1/2}\mu^{1/2}} \tag{20}$$

where dT/dR_W is the average temperature gradient in the weld pool. Using Eq. (20), we estimate the maximum velocity to be of the order of 1.433 m/s for the sample 1. This value is in good agreement with that calculated using the 3D transient heat transfer and fluid flow model, where the maximum velocity was found to be about 1.445 m/s.

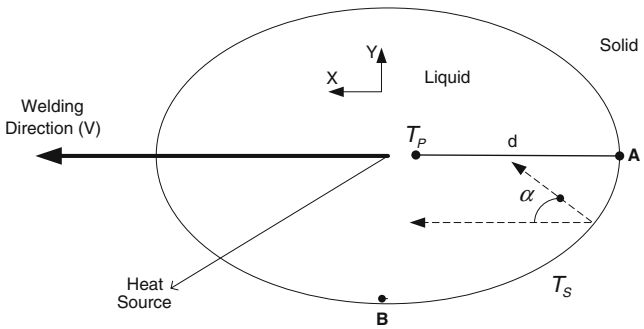


Fig. 9 A schematic plot showing the location of heat source, peak temperature, and the calculation of average temperature gradient in the weld pool

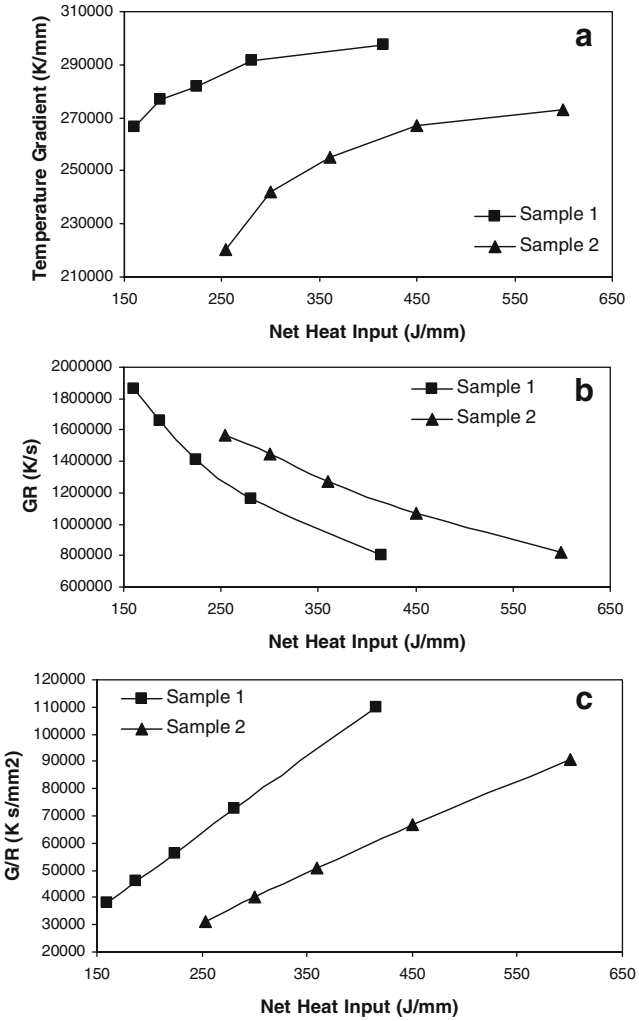


Fig. 10 Calculated values of (**a**) G, (**b**) GR and (**c**) G/R at the weld pool centerline with different net heat input for the samples 1 and 2

The foregoing dimensional analysis provided insights about the weld pool development. The role of conduction and convection on heat transfer in the weld pool, the relative importance of driving forces for liquid metal convection, the order of magnitude of the velocity driven by the surface tension force, and the weld pool shape could be estimated using several dimensionless numbers. It should be noted that these order of magnitude analyses cannot provide accurate and detailed information on the welding processes, which requires numerical calculation with a very fine grid system.

4.4 Solidification

As shown in Fig. 9, the average temperature gradient, $\overline{G}_a$, of the weld pool along a given plane is defined as

$$\overline{G}_a = \frac{T_P - T_S}{d} \tag{21}$$

where T_P is the peak temperature, T_S is the solidus temperature, and d is the distance between the location of peak temperature and boundary at the weld pool centerline.

Since the shape of the weld pool remains constant under steady state condition during welding, the solidification rate varies with position along the fusion boundary. This is illustrated in Fig. 9, which shows that the direction of movement of the solidification front is along the maximum thermal gradient normal to the solid/liquid interface. Therefore, the steady state solidification rate, R, is related to the welding speed as [36]

$$R = V \cos \alpha \tag{22}$$

where α is the angle between the normal to the solid/liquid interface and the welding direction and V is the welding speed. The solidification rate is lowest at the edge of the weld pool ($\alpha \rightarrow 90°$), indicated by point B in Fig. 9. The solidification rate is highest at point A on the weld centerline because the interface normal has the same direction as the welding direction. The solidification rate decreases form point A to point B along the fusion boundary. The temperature gradient and the solidification rate are important in the combined forms G/R and GR as they influence the solidification morphology and the scale of the solidification substructure, respectively. As the value of G/R increases, the interface morphology changes from equiaxed-dendritic, to cellular-dendritic, to cellular grains [36]. Often the value of G/R close to the fusion line is large enough to facilitate cellular solidification. When the temperature gradient in the weld pool decreases with distance from the fusion boundary, the solidification microstructure may vary spatially. For example, in some cases, an equiaxed-dendritic microstructure at the weld pool center and a cellular-dendritic microstructure between fusion boundary and weld centerline may be obtained. Figure 10 shows the temperature gradient G and G/R at the weld centerline for different net heat inputs that were simulated by varying the welding speed for the samples 1 and 2. It is shown that with the increase in net heat input or as a decrease in welding speed, the temperature gradient increases, the value of GR decreases, and the value of G/R increases at the weld pool centerline. The similar behavior is shown in the literature [37]. Thus the solidification structure of fusion zone would be affected by the net heat input.

5 Conclusions

The temperature and velocity fields, weld thermal cycles, weld pool size, and the solidification parameters during GTA welding of commercial pure aluminum were simulated using a comprehensive three-dimensional heat transfer and fluid flow model. The major conclusions can be made from investigation are as follows:

(1) The geometry of fusion zone predicted from the three-dimensional steady state heat transfer and fluid flow model using FLUENT code are in a good agreement with the corresponding experimental results.
(2) The peak temperature increases but heating and cooling rates decreases with decreasing the welding speed. Also heating rates is higher than cooling rates.
(3) The dimensional analysis show that liquid metal convection is the dominate mechanism of heat transfer and determines the temperature distribution in the liquid pool and the pool shape. In addition, the role of convection becomes more important as the net heat input increases or the welding speed decreases.
(4) The value of G/R at the weld pool centerline is shown to increase with the increase in net heat input and decrease in welding speed.

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